

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403303
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Chavez; Alfonso et al.

Lead fixation devices for securing a lead to a cranium

Abstract

A lead fixation device for securing a first portion of a lead relative to a hole formed through a skull includes a skull attachment member having an upper surface and a lower surface and a bore extending through and between the upper surface and the lower surface. The lead fixation device also includes a lead compression mechanism integral with the skull attachment member and aligned with the bore of the skull attachment member. The lead compression mechanism defines a passageway through the lead fixation device, which passageway is characterized by a diameter that is defined by the lead compression mechanism. The lead compression mechanism is configured to transition the diameter from a first size to a second size greater than the first size upon insertion of an implant tool through the passageway, and from the second size to the first size upon removal of the implant tool from the passageway.

Inventors:	Chavez; Alfonso (San Jose, CA), Greene; David A. (Fort Wayne, IN)
Applicant:	NeuroPace, Inc. (Mountain View, CA)
Family ID:	70851060
Assignee:	NeuroPace, Inc. (Mountain View, CA)
Appl. No.:	18/077668
Filed:	December 08, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230104746 A1	Apr. 06, 2023

Related U.S. Application Data

division parent-doc US 16692964 20191122 US 11547850 child-doc US 18077668
us-provisional-application US 62773121 20181129

Publication Classification

Int. Cl.: A61N1/00 (20060101); A61N1/05 (20060101)

U.S. Cl.:

CPC A61N1/0539 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4328813	12/1981	Ray	N/A	N/A
5464446	12/1994	Dreesen et al.	N/A	N/A
5843150	12/1997	Dressen et al.	N/A	N/A
5865842	12/1998	Knuth	607/116	A61B 5/24
5865843	12/1998	Baudino	N/A	N/A
5927277	12/1998	Baudino et al.	N/A	N/A
6044304	12/1999	Baudino	N/A	N/A
6134477	12/1999	Knuteson et al.	N/A	N/A
6210417	12/2000	Baudino et al.	N/A	N/A
6319241	12/2000	King et al.	N/A	N/A
6321104	12/2000	Gielen et al.	N/A	N/A
6356792	12/2001	Errico et al.	N/A	N/A
6795737	12/2003	Gielen et al.	N/A	N/A
6810285	12/2003	Pless et al.	N/A	N/A
6902569	12/2004	Parmer et al.	N/A	N/A
7204840	12/2006	Skakoon et al.	N/A	N/A
7235084	12/2006	Skakoon et al.	N/A	N/A
7580756	12/2008	Schulte et al.	N/A	N/A
7604644	12/2008	Schulte et al.	N/A	N/A
7636596	12/2008	Solar	N/A	N/A
7637915	12/2008	Parmer et al.	N/A	N/A
7660621	12/2009	Skakoon et al.	N/A	N/A
7704260	12/2009	Skakoon et al.	N/A	N/A
7744606	12/2009	Miller et al.	N/A	N/A
7766394	12/2009	Sage et al.	N/A	N/A
7766922	12/2009	Daglow et al.	N/A	N/A
7815651	12/2009	Skakoon et al.	N/A	N/A
7828809	12/2009	Skakoon et al.	N/A	N/A
7833231	12/2009	Skakoon et al.	N/A	N/A
7857820	12/2009	Skakoon et al.	N/A	N/A
7949410	12/2010	Rodriguez	N/A	N/A
7976530	12/2010	Johnson et al.	N/A	N/A
7988674	12/2010	Adams et al.	N/A	N/A

8116850	12/2011	Solar	N/A	N/A
8152792	12/2011	Kornel	N/A	N/A
8192445	12/2011	Parmer et al.	N/A	N/A
8845656	12/2013	Skakoon et al.	N/A	N/A
8911452	12/2013	Skakoon et al.	N/A	N/A
9457180	12/2015	Bucholz	N/A	N/A
9545509	12/2016	Greene	N/A	N/A
9572973	12/2016	Chavez et al.	N/A	N/A
2005/0015128	12/2004	Rezai et al.	N/A	N/A
2005/0182421	12/2004	Schulte et al.	N/A	N/A
2005/0182422	12/2004	Schulte et al.	N/A	N/A
2005/0182424	12/2004	Schulte et al.	N/A	N/A
2005/0182425	12/2004	Schulte et al.	N/A	N/A
2009/0112327	12/2008	Lane et al.	N/A	N/A
2009/0306750	12/2008	Boling et al.	N/A	N/A
2009/0326610	12/2008	Pless et al.	N/A	N/A
2010/0179563	12/2009	Skakoon et al.	N/A	N/A
2010/0312193	12/2009	Stratton	N/A	N/A
2011/0270187	12/2010	Nelson	N/A	N/A
2012/0245529	12/2011	Hummen et al.	N/A	N/A
2013/0066410	12/2012	Funderburk	N/A	N/A
2013/0304216	12/2012	Papsa et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2001176498	12/2000	WO	N/A
2004103468	12/2013	WO	N/A

OTHER PUBLICATIONS

Biomet Microfixation. Neuroimplant System. Product Brochure, 8 pgs. (2013). cited by applicant
Aesculap Inc. Cranial Fixation Systems. Product Brochure, 11 pgs. (2014). cited by applicant
Miller et al. "Stereotactic bony trajectory preservation for responsive neurostimulator lead placement following depth EEG recording." Cureus 8(3):3549. DOI 10.7759/cureus.549, 6 pgs. (Mar. 30, 2016). cited by applicant
Synthes CMF. "MatrixNEURO. The next generation cranial plating system." Product Brochure, 10 pgs. (2006). cited by applicant

Primary Examiner: Downey; John R

Assistant Examiner: Gupta; Anant A

Attorney, Agent or Firm: LOZA & LOZA, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/692,964, filed on Nov. 22, 2019, now U.S. Pat. No. 11,547,850, entitled "Lead Fixation Devices for Securing a Lead to a Cranium," which claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 62/773,121, filed Nov. 29, 2018, for "Lead Fixation

Devices for Securing a Lead to a Cranium,” the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to apparatuses and methods used when implanting a medical device in a patient, and more particularly, to lead fixation devices for securing a lead to a cranium and methods for using the same.

BACKGROUND

(2) Some diagnostic or interventional medical procedures require implanting one or more leads through a hole in a patient's cranium. Once the surgeon places a lead so that the distal end is at a desired location, the surgeon wants the lead to stay there for whatever ultimate medical purpose the lead has, for example, to monitor physiological parameters from a patient or to deliver a therapy to the patient. Usually, however, there are steps subsequent to placing the lead that are necessary to complete the surgical procedure, and some of these may cause inadvertent displacement of the lead away from the desired location, which may not be corrected easily or efficiently.

(3) Generally, leads may be provided with electrodes configured to sense information from the brain or to deliver a form of stimulation to the brain intended to modulate neural activity, such as electrical stimulation. The sensing and/or stimulation may occur at a distal end of the lead, for example, through electrodes exposed to brain tissue at a distal end, wherein the signals are communicated through conductors disposed in the lead body extending to a lead proximal end. Connections available at the lead proximal end allow the lead to be connected to another medical device, implanted or external, that processes the sensed signals and/or generates the form of stimulation.

(4) For example, in the responsive neurostimulation system manufactured under the tradename RNS SYSTEM by NeuroPace, Inc., the proximal ends of one or more implanted brain leads can be connected to another implanted medical device, namely a neurostimulator that is seated in a tray or ferrule in a craniectomy in the patient's skull. In another example, the leads may be implanted in a patient so that an intracranial monitoring procedure can be undertaken for a period (e.g., several days or a couple of weeks), with the proximal ends of the implanted leads being connected to external equipment monitoring the patient's brain activity, such as to identify a focus or the foci of epileptiform activity in the patient.

(5) There are multiple types of brain leads currently available. In applications where the leads are being used to sense or stimulate brain tissue at or near the focus of undesirable epileptiform activity, there is a depth lead (also sometimes referred to as a “deep brain lead” or as a “stereotactic depth lead”, because this lead type is often implanted using stereotaxy, a three-dimensional localization and placement procedure) and a cortical strip lead (also known simply as a “cortical lead” or as a “subdural lead”, because this lead type is usually implanted underneath the dura mater).

(6) A depth lead is implanted so that the distal end is located in the brain tissue, in or adjacent a structure that is deemed to be associated with the generation of the undesirable activity. A cortical strip lead is implanted so that the distal end lays on a surface of the brain at or adjacent brain tissue that is believed to comprise an epileptic focus. The intended location of the distal end of the brain lead in or on the brain is referred to hereinafter as the “target”.

(7) A lead manufacturer may make different lead types in one or more standard lengths, rather than in lengths customized for a particular application in a particular patient. In addition, the depth lead type may be intended to be implanted using stereotactic equipment or some other tool that requires some length in excess of that which is needed to extend from the cranium and the connection to another medical device and the target (e.g., so that the lead is long enough to extend through the distance required when using a stereotactic frame mounted to the patient). For at least the reason that a lead may be manufactured to have more length than is necessary to traverse the distance

between the proximal connection to another implant or external equipment and the target, brain leads are often manufactured to be quite flexible. That is, if the lead is flexible, excess length can be coiled or folded at the surface of the skull before the scalp is replaced. Further, flexibility may be considered a better alternative than a stiff lead when the lead is to remain in place in or on a surface of the brain chronically, as opposed to acutely, such as to minimize tissue damage and to optimize the integrity or resolution of signals. Brain leads manufactured and sold with the RNS SYSTEM, for example, have a flexibility on par with that of a piece of cooked spaghetti.

(8) When implanting a lead in the brain tissue, though, its relative flexibility can present challenges when delivering the distal end to the target. Accordingly, a brain lead is often provided with an inner lumen through which a removable stiffener, such as a stylet, can be disposed. The stylet lends stability to the lead while the distal end is routed to the target, and is then removed when the lead has been positioned where it is intended to remain, either acutely or chronically. It is undesirable, however, if the act of withdrawing the stylet causes the distal end to move away from the target.

(9) Equipment or tools used in implanting a brain lead can also unintentionally cause the distal end of a brain lead to move away from the target in procedural steps undertaken subsequent to placing the lead. For example, a slotted cannula is often employed in implanting leads stereotactically, in which a hole is formed in the patient's skull at a location calculated to allow an appropriate trajectory of a lead to a deep brain target. In one example of such a procedure, a frame is attached to the patient, and a guide tube is oriented to achieve the desired trajectory relative to the skull hole. A cannula is inserted through the guide tube, such that its range of motion is constrained by the guide tube. The cannula may be provided with a removable rod disposed in an inner lumen thereof. The cannula is advanced through the skull hole towards the target. The inner rod in the cannula prevents tissue from backing up into the cannula while it is advanced.

(10) When the cannula has been advanced to or approximate the target, the inner rod is removed and a depth lead inserted into the cannula lumen. The surgeon then advances the depth lead to the target. (Sometimes the depth lead is marked in advance at a proximal location, for example, with a stop gauge, to provide feedback to the surgeon when the target has been reached.) Once the distal end of the lead reaches the target, the cannula must be withdrawn from the brain and the lead must be extracted from the cannula, so the stereotactic equipment can be removed. A cannula is often provided with a longitudinally-extending slot with a width wide enough to accommodate the diameter of the lead body for this purpose, i.e., so that the lead can be stripped away from the cannula using the slot, rather than having to retract the cannula over the lead body. This allows the lead to be manufactured with somewhat less excess lead length than if the lead had to be long enough to retract the cannula over the very proximal end of the lead. As is the case with withdrawing a stylet, it is undesirable if the act of disengaging the lead body from a cannula (or other apparatus used for stereotaxy) causes the distal end of the brain lead to move away from the target.

SUMMARY

(11) The present disclosure relates to a lead fixation device for securing a first portion of a lead relative to a hole formed through a skull. The lead fixation device includes a skull attachment member having an upper surface and a lower surface and a bore extending through and between the upper surface and the lower surface. The lead fixation device also includes a lead compression mechanism that is integral with the skull attachment member and aligned with the bore of the skull attachment member. The lead compression mechanism defines or forms a passageway through the lead fixation device, which passageway is characterized by a diameter that is defined by the lead compression mechanism. The lead compression mechanism is configured to transition the diameter from a first size to a second size greater than the first size upon insertion of an implant tool through the passageway, and from the second size to the first size upon removal of the implant tool from the passageway.

(12) In one configuration, the compression mechanism is fixedly secure relative to the skull

attachment member so that transition of the diameter between the first size and the second size results from radial outward compression of the compression mechanism and radial inward expansion of the compression mechanism relative to an axis through the passageway. The first portion of the lead has a diameter and the first size of the diameter of the compression mechanism is less than the diameter of the lead to thereby secure the first portion of the lead in place at the skull hole.

(13) The present disclosure also relates to a lead fixation device for securing a portion of a lead relative to a surface of a skull. The lead fixation device includes a skull attachment member having an upper surface and a lower surface. The lead fixation device also includes a lead compression mechanism that is integral with the skull attachment member. The lead compression mechanism is formed of a flexible material while the skull attachment member is formed of a material more rigid than the lead compression mechanism. The lead compression mechanism forms a passageway through the skull attachment member. In one embodiment, the passageway is characterized by a slot that faces in the direction of the lower surface and extends along the length of the passageway. In another embodiment, the passageway is characterized by a slot that faces in the direction of the upper surface and extends along the length of the passageway.

(14) The lead compression mechanism is formed of a flexible material and is configured to transition a width of the slot from an initial size that is less than the diameter of the portion of the lead, to an expanded size that is greater than the initial size upon receipt of appropriately directed first forces in a region of the lead compression mechanism that forms the passageway. Such first forces increase the width and allow for the portion of the lead to slide into the passageway. The lead compression mechanism is further configured to transition the width of the slot from the initial size to a collapsed size upon receipt of appropriately directed second force in a region of the lead compression mechanism that forms the passageway. Such second forces decrease the width and secure the portion of the lead inside the passageway.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of apparatuses and methods will now be presented in the detailed description by way of example, and not by way of limitation, with reference to the accompanying drawings, wherein:

(2) FIG. 1A is a schematic of a patient's cranium showing implanted components of a neurostimulation system, including leads and a neurostimulator, and related surgical accessories, including a burr hole cover and a lead fixation plate.

(3) FIGS. 1B and 1C are illustrations of known burr hole covers.

(4) FIG. 1D is an illustration of a known lead fixation plate.

(5) FIG. 2 is an illustration of some components of stereotactic equipment that may be used in a standard stereotactic procedure with a frame to implant a depth lead in a patient's brain.

(6) FIG. 3 is an illustration of a cannula (with a depth lead inserted therein) that may be used during a procedure for implanting a depth lead.

(7) FIGS. 4A-4D are illustrations of a first embodiment of a lead fixation device for securing a lead at a hole in a skull and on a surface of the skull.

(8) FIGS. 5A-5C are illustrations of a skull attachment member of the lead fixation device of FIGS. 4A-4D.

(9) FIG. 6 is a side view drawing of the skull attachment member of FIG. 4A-4D.

(10) FIGS. 7A and 7B are illustrations of a compression mechanism of the lead fixation device of FIGS. 4A-4D.

(11) FIGS. 8A and 8B are illustrations of a locking member of the lead fixation device of FIGS.

4A-4D, that secures the compression mechanism in place in the skull attachment member.

(12) FIGS. **9A** and **9B** are illustrations of a second embodiment of a lead fixation device for securing a lead at a hole in a skull and on a surface of the skull.

(13) FIG. **10A-10J** are illustrations of a lead implant procedure using the lead fixation device of FIGS. **4A-4D**.

(14) FIGS. **11** and **12** are illustrations of a first embodiment of a lead fixation device for securing a lead on a surface of the skull.

(15) FIG. **13** is an illustration of a lead implant procedure using the lead fixation device of FIGS. **11** and **12**.

(16) FIGS. **14**, **15A**, and **15B** are illustrations of a second embodiment of a lead fixation device for securing a lead on a surface of the skull.

(17) FIGS. **16A** and **16B** are illustrations of a lead implant procedure using the lead fixation device of FIG. **14**.

DETAILED DESCRIPTION

(18) Disclosed herein is a lead fixation device that allows the body of a brain lead to be secured relative to a hole formed through the skull and at the surface of a patient's skull while a stiffening element, such as a stylet, remains in place in a lumen of the lead body. The stiffening element thus can be withdrawn from the lead after the lead fixation accessory is in place, reducing the likelihood that the act of pulling out the stylet will displace the distal end of the implanted lead away from the target. Also disclosed herein are embodiments of lead fixation device that allow the body of a brain lead to be secured at or along a surface of a skull at a location between the point where the lead body exits the skull and the point where the lead connects to an implanted medical device.

(19) The embodiments are described primarily with reference to the lead being an electrode-bearing lead, as might be used in an application for deep brain stimulation or direct brain stimulation such as the responsive stimulations applications by NeuroPace, Inc. of Mountain View, CA. It should be appreciated, however, that the lead fixation devices may be used with good results to secure a segment of a different type of medical device, such as a catheter or other medical instrument (with a diameter compatible with the accessory), relative to a surface of the skull prior to and/or during use of the medical device in its intended application.

(20) Overview of Procedures to Implant Brain Leads

(21) For purposes of illustration, procedures to implant a depth lead and a cortical strip lead will be described with reference to a responsive neurostimulation system, in which a surgeon commonly uses both lead types.

(22) With reference to FIG. **1A**, a neurostimulator **110** and leads **120**, **130** of a responsive neurostimulation system are shown schematically, implanted in a patient. To implant a lead, the surgeon needs access to the brain. A surgeon may gain access to the brain for purposes of implanting a lead by creating an opening through the skull. A opening may be created by drilling a hole through the skull, by performing a craniotomy (temporarily removing a bone flap from the skull and replacing the flap after access to the brain is no longer needed) or by performing a craniectomy (permanently removing a bone flap from the skull). Such opening may be used exclusively for lead implant purposes, or may be used for another/additional purpose (for example, the surgeon can first deliver a lead to a target through an opening formed as part of a craniectomy, then use the same opening to implant another medical device, such as a neurostimulator). The term "skull hole" is used herein to refer to any category of opening formed in a patient's skull to gain access to the subdural spaces and to the brain.

(23) In FIG. **1A**, three skull holes have been formed: a burr hole **140** for purposes of implanting a depth lead **120**, a craniotomy **150** for purposes of implanting a cortical strip lead **130**, and a craniectomy **160** in which a ferrule or tray **112** and a neurostimulator **110** are ultimately implanted. More particularly, the surgeon may use an air-powered drill to form an annular burr hole **140** of a diameter between 5-30 mm, with 14 mm being a commonly-used diameter, for purposes of

implanting a depth lead **120**. Using appropriate tools, the surgeon may also perform a craniotomy **150** for purposes of implanting a cortical strip lead **130**, and additionally a craniectomy **160** in which to ultimately situate a neurostimulator at the patient's skull.

(24) In FIG. **1A**, a distal portion **122** of the depth lead **120** extends into the patient's brain tissue from a 14-mm burr hole **140**, and a proximal portion **128** extends proximally from the burr hole where it is plugged in at a proximal end **129** to a connector **114** of an implanted neurostimulator **110**. A distal portion **132** of a cortical strip lead **130** extends from a fissure like hole or opening **151** at an edge of the craniotomy **150** onto a surface of the patient's brain, between the brain and the dura mater (not shown), and a proximal portion **138** extends proximally from the hole where it is plugged in at a proximal end **139** to the connector **114**. The neurostimulator **110** has a strain relief **116** in the location where the proximal ends **129**, **139** of the leads connect, to discourage the leads from unintentional disconnection.

(25) A distal end **124** of the depth lead **120** includes a plurality of electrodes **126** (three are shown in FIG. **1A**), that can be used either for sensing electrographic activity from the brain or for delivering a therapy of electrical stimulation to it in an effort to modulate neural activity (e.g., lessen the severity of a seizure). Conductors extending the length of the lead body (not shown) and connected at the connector **114** to the neurostimulator **110** allow the neurostimulator to process the sensed signals and to generate the stimulation signals. A distal end **134** of the cortical strip lead **130** ends in a paddle **136** that, on a brain-facing surface thereof (not shown in FIG. **1A**), exposes another plurality of electrodes (e.g., four) to the brain surface underneath the dura mater. These electrodes are also in electrical communication with the neurostimulator **110** via conductors in the cortical strip lead **130** and the connection at the connector **114**.

(26) In addition to the burr hole **140** or the craniotomy **150** opening **151**, a lead, especially of the cortical strip lead type, may be implanted using another opening in the cranium. More specifically, to implant the neurostimulator **110**, the surgeon cuts a craniectomy **160** hole using a template that approximates the shape of the neurostimulator. The surgeon fits a tray or "ferrule" **112** into the hole and attaches or otherwise secures it to the cranium, for example, using bone screws and/or folding tabs **118** providing on the tray. The surgeon then situates the neurostimulator **110** into the tray **112**. However, before placing the tray **112**, the surgeon can use the craniectomy **160** hole to implant a cortical strip lead, such as the cortical strip lead **130**, and then connect the proximal end thereof to the neurostimulator connector. (FIG. **1A** does not show any lead implanted using the craniectomy **160** in which the tray **112** and neurostimulator **110** are situated.)

(27) Both of the implanted leads **120**, **130** in FIG. **1A** are shown secured with known lead fixation accessories. The depth lead **120** implanted through the burr hole **140** is shown secured with a burr hole cover **144** which substantially fills the 14-mm diameter burr hole except for an aperture therethrough that permits passage of the lead body. Examples of lead fixation accessories designed for burr holes are illustrated in FIGS. **1B** and **1C**.

(28) Some burr hole lead fixation devices are designed for use with mechanical parts that need to be actuated in order to achieve fixation of the lead body, and others rely on friction fit or compression to limit movement of the lead relative to the device. Some require at least one element of the accessory to be put in place before a procedure to implant a lead is begun. Some allow fixation only after any stiffening element used in implanting the lead has been removed. With reference to FIG. **1B**, a burr hole lead fixation accessory manufactured by Medtronic, Inc. under the tradename "STIMLOC" uses several interlocking parts to secure a lead body. With reference to FIG. **1C**, a two-piece burr hole cover manufactured by NeuroPace, Inc. relies in part on fitting a portion of the lead body into a groove in base element to reduce the likelihood that further manipulation of the lead portion extending proximally of the skull hole (e.g., to connect the lead to an implanted neurostimulator) will translate to movement of the distal end away from the target.

(29) In FIG. **1A**, the cortical strip lead **130** implanted through the craniotomy **150**, and having a proximal portion **138** extending from the opening **151** at an edge of the craniotomy onto the surface

of the skull is secured at a point on the lead body just proximal of where the lead body extends out of the hole, with a cranial plate **154**. The cranial plate **154** is situated over the lead body and then secured to the surface of the skull on either side of the lead body with bone screws. Because of its shape, the type of cranial plate **154** shown in FIG. **1A** is commonly referred to as a “dog bone”. One such plate is shown in FIGS. **1D** and **1s** manufactured under the tradename “MATRIXNEURO” by Synthes CMF. The cranial plate **154** compresses the lead body to prevent lateral movement of the lead at the point of fixation to the skull. If the compression is inadvertently excessive (e.g., by overtightening of the screws or by a patient pressing down on the plate), the integrity of the lead may be compromised (e.g., the conductors between the electrodes at the lead distal end and the connector at the lead proximal end may be shorted).

(30) The target for a depth lead **120** is usually more precise than the target for a cortical strip lead **130**, at least in an application where the condition is epilepsy. That is, the target for a depth lead **120** is usually a particular structure in the brain, such as the subthalamic nucleus (STN) or the cingulate gyrus. The target for a cortical strip lead **130** may be somewhat more forgiving of imprecision, that is, the electrodes on the distal end **124** of the strip lead **130** may be destined to cover the general area on the surface of the brain where epileptic activity is believed to be focused. Thus, it may be especially beneficial to limit movement of the distal end of a depth lead once it has been placed at the target.

(31) In part because of the need for precision and in part because the lead is being implanted into brain tissue as opposed to on a surface of it, a depth lead **120** is most often implanted using some form of stereotaxy (e.g., with a frame affixed to the patient's skull or a “frameless” version of it). Stereotactic procedures are well known and will not be described herein to any great degree. Briefly, however, one common method uses frame-based stereotaxis to approach a target or targets through a skull hole. The patient is given a local anesthetic and a rigid frame or fixation device is attached to the patient's head, and the brain is imaged (e.g., with a CT scan). The location of the target(s) is calculated based on a ‘co-registration’ of the images and the frame, fiducials or other registered points on the head. Then, the patient is sedated for surgery, the scalp is incised, and one or more skull holes are formed in the patient's cranium, each at a location that will allow an appropriate trajectory to the deep brain target(s).

(32) A hole in the skull is often formed with some standard diameter, owing to the drills typically available in the operating room to create it. When an air drill is used to create a hole in the skull with a diameter of 5 mm or greater, the skull hole is often referred to as a “burr hole.” Surgeons create standard-sized burr holes, because there are surgical accessories intended for use with burr holes that are intended for use with certain burr hole diameters, such as 14 mm. However, the diameter of a brain lead may be much smaller than that of a burr hole, because 14 mm is on the order of ten times greater than the diameter of the lead to be implanted. For example, some brain leads manufactured by NeuroPace, Inc. have a diameter of only 1.27 mm. Therefore, in some cases a surgeon may choose to use a smaller diameter hole through which to implant a lead. For example, a surgeon may choose to use a hand-held twist drill to create a hole with a diameter on the order of less than 5 mm (depending on the diameter of the twist drill bit: a common one results in a 3.2 mm diameter hole). A skull hole formed using a twist drill is sometimes referred to as a “twist drill hole”.

(33) Referring now to FIGS. **2** and **3**, part of the stereotactic equipment **200** is secured to the patient's skull using a frame, a portion of which is shown as a graduated element **202** in FIG. **2**, and a guide tube **204** is oriented to provide the desired trajectory. The guide tube **204** has an inner lumen of sufficient diameter to receive a cannula **304**. The cannula **304** is also formed as a cylinder, typically made of a metal, such as stainless steel, and has an inner lumen with a diameter sufficient to slidably receive first an inner rod (not shown) and thereafter a depth lead (the distal end **324** of a depth lead **320** is shown in FIG. **3**).

(34) The cannula **304** may be provided with a slot **306** running along its entire length so that the

lead body can be extracted from the cannula without having to retract the cannula over the proximal end **330** of the lead. Thus, if the cannula **304** is slotted, the slot **306** must be dimensioned so as to allow the proximal portion of the depth lead **320** that extends proximally of the skull hole to be separated from the cannula through the slot. The depth lead **320** also has an inner lumen running through most of the length of the lead in which a stiffening element, such as a stylet, is removably disposed. (In FIG. 3, a stylet **302** is shown extending proximally of the depth lead **320**). The stylet **302** may have a plastic member or stylet handle **303** at its proximal end that the surgeon can grab to more easily extract the stylet from the lead inner lumen.

(35) One or more stop gauges may be configured so that they can encircle the proximal portion of either or both of the cannula **304** or the depth lead **320** to guard against advancing the distal end of the cannula or the depth lead beyond the target (not shown in FIG. 2 or 3). For example, the depth lead **320** may be measured in the operating room to identify a location on a proximal portion that, once the lead is being routed to the target, the surgeon can use to gauge when the lead has been advanced far enough (or to some not-to-exceed distance) into the tissue. This location on the proximal portion can be demarcated by fitting a stop gauge **310** around the lead body.

(36) Manipulating the appropriate controls on the stereotactic equipment, the cannula **304** with the inner rod (not shown) in place is advanced into the brain. The inner rod discourages brain tissue from backing up into the cannula lumen as the cannula creates a path to the target for the lead. When the cannula **304** is advanced as far as intended, the surgeon withdraws the inner rod, and replaces it with the depth lead **320**, by inserting the distal end **324** of the depth lead (with the stylet **302** in place) into the proximal end **312** (or top) of the cannula.

(37) FIG. 3 shows a cannula **304** with a depth lead **320** inserted within the cannula inner lumen. A proximal portion **328** of the depth lead **320** extends proximally of a proximal end **312** of the cannula **304**, and a distal portion **322** of the depth lead extends distally of a distal end **308** of the cannula **304**. The stylet **302** is disposed in an inner lumen of the depth lead **320** and traverses substantially the full length of the depth lead **320**, except for the very distal end **324** thereof. The stylet **302** is shown extending proximally of the proximal end **330** of the depth lead **320**, with a stylet handle **303** at the proximal tip. The stylet **302** lends sufficient stiffness to the lead **320** while it is being manipulated during the implant procedure (e.g., to insert it into the cannula lumen. The stylet handle **303** makes it easier to remove the stylet **302** from the lead **320** before the procedure is over. It will be appreciated that in a typical stereotactic procedure, even when the depth lead **320** is inserted into the cannula **304** and after the lead distal end **324** has been delivered to the target, there is enough excess lead length so that a portion of the lead body will extend proximally of the proximal end **312** of the cannula, so that the lead at a point on the proximal portion **328** thereof can be grasped above the proximal end **312** of the cannula **304**.

(38) After the step in the procedure where the surgeon has the distal end **324** of the lead **320** where he or she wants it, the cannula **304** is removed while the lead is separated from the cannula through the slot **306**. After the cannula **304** is removed, it is undesirable for subsequent steps to move the distal end **324** of the lead away from the target. But preventing that from happening can be challenging because, after the cannula **304** is removed, the stylet **302** in the inner lumen of the lead **320** still has to be extracted from the lead body before the procedure is complete. The force applied in pulling out the stylet **302** may tend to retract the distal end **324** of the lead along with it, so removing the stylet is another step which may result in dislodging the lead away from the target.

(39) Furthermore, some form of lead fixation device typically is used to secure a proximal portion of the implanted lead at or near the skull hole or otherwise somewhere on the surface of the skull, to discourage relative movement between the implanted distal portion of the lead and the proximal portion of the lead after the procedure is complete. The step is another opportunity for unwanted displacement of the distal end of the lead from the target.

(40) Lead Fixation Device with Compression Mechanism at Skull Hole

(41) With reference to FIGS. 4A-9B, described are embodiments of a lead fixation device **400**, **900**

configured to compress against a lead body to thereby secure the lead body relative to a skull hole and prevent or at least significantly reduce movement of the lead body either further into the skull hole or out of the skull hole during implant of the lead. This lead fixation device **400, 900** secures a first portion of the lead at a skull hole where the part of the lead distal the first portion extends through the skull hole in a direction transverse to the skull surface, and secures a second portion of the lead relative to a skull surface such that the second portion rests on and along a skull surface, in a parallel arrangement with the skull surface. The lead fixation device **400, 900** overcomes the lead implanting issues described above with reference to FIGS. 2 and 3, including in particular the issues of lead movement during removal of a cannula **304** from around a lead **320** and removal of a stylet **302** from a lead.

(42) The lead fixation device **400, 900** includes a skull attachment member **402, 902** configured to be secured to the skull and a flexible compression mechanism **406, 906** that is associated with the skull attachment member. The compression mechanism **406, 906** is configured to compress radially outward to expand its inner diameter to receive a lead implant tool, e.g., a cannula, and then expand radially inward to reduce its inner diameter to compress against a portion of the lead body upon removal of the lead implant tool.

(43) With continued reference to FIGS. 4A-9B, the skull attachment member **402, 902** includes a central bore **410, 910** and lead securing features corresponding to surface channels **414, 914** formed in the body **415, 915** of the of the attachment member and that extend from the central bore to the outer perimeter **417, 917** of the attachment member. The skull attachment member **402, 902** may also include a pair of opposed protrusions **416, 916** at a location along the length of each of the surface channels **414, 914**. These protrusions **416, 916** extend inward toward the center **419, 919** of the channel **414, 914** to provide an area of reduced width of the channel that functions to secure a portion of a lead body in place in the channel. The skull attachment member **402** also includes a pair of screw holes **412, 912** for receiving bone screws **408, 908** that secure the lead fixation device **400, 900** to the skull. The bone screws **408, 908** may be preinstalled into the screw holes **412, 912** via an interference fit, or they may be separately installed through the holes after placement of the skull attachment member **402, 902**.

(44) With reference to FIGS. 6-8B, the compression mechanism **406** is located in a first recess **418** formed in the skull attachment member **402** that surrounds the central bore **410**, and is secured in place by a locking ring **404** that fits in a second recess **420** that also surrounds the central bore. The skull attachment member **402** and the locking ring **404** are formed of a rigid plastic, e.g., PEEK. The locking ring **404** includes a circular recess **430** sized to receive the compression mechanism **406**. The diameter **432** of the recess **430** is slightly greater than the outer diameter **426** of the compression mechanism to provide a space for radial expansion of the compression mechanism and a corresponding increase in the size of its inner diameter **424** when a cannula is inserted through the opening defined by the compression mechanism.

(45) The compression mechanism **406** is formed of a flexible material, e.g., silicone, and is configured such that the inner diameter **424** defined by the compression mechanism expands to accommodate insertion of a cannula **304** and contracts to fit tightly around a lead body (when the cannula is removed from the compression mechanism) to prevent movement of the lead relative to a skull hole. Properties of the compression mechanism, including the inner diameter **424**, the outer diameter **426** and the cross-section diameter **428**, and the stiffness of the compression mechanism material may be selected to obtain the desired results.

(46) While the compression mechanism **406, 906** in the embodiments in FIGS. 4A-9B is in the form of an O-ring, other types of compression mechanisms are contemplated. For example, the compression mechanism may be defined by a slot with silicone on each side that defines a cross-section width that compresses a portion of a lead body sufficiently to secure the lead in place in relative to a skull hole.

(47) With reference to FIGS. 10A-10J, an example implant procedure for the lead fixation device

400 of FIGS. **4A-8B** is provided. A twist drill hole **440** is formed in a location of the cranium **444** using known stereotactic techniques. (See FIG. **10A**.) The distal end of a split (or slotted) cannula **304** (with an inner rod **446** inserted) is slid through a passageway or bore **410** of the lead fixation device **400** that is defined by a compression mechanism **406** of the device. The size of the passageway, which is defined by an expanded inner diameter **424** of the compression mechanism **406**, exerts sufficient radial pressure on the cannula **304** to hold the lead fixation device **400** in place on the cannula. (See FIG. **10B**.)

(48) The distal portion of slotted cannula **304** (with the inner rod **446** inserted) it is inserted through the twist drill hole **440**, into the brain to a known depth using stereotactic techniques. (See FIG. **10C**.) The lead fixation device **400** is slid down the cannula **304** in direction A (see FIG. **10c**) to rest on top of the cranium **444** and is secured in place using bone screws **408**. (See FIG. **10D**.)

(49) The inner rod **446** in the cannula **304** is removed by pulling on the inner rod **446** in direction B. (See FIG. **10E**.) The lead **320** with a stylet **302** inserted is inserted into the cannula **304** and advanced in direction C to the known/predetermined depth using known stereotactic techniques. (See FIG. **10F**.)

(50) The cannula **304** is removed from around the lead **320** through the slot **306** of the cannula. (See FIG. **10G**.) Or, in other words, the lead **320** is separated from the cannula **304** through the slot **306**. When the cannula **304** is removed, the size of the passageway of the lead fixation device **400**, which is defined by a slightly less expanded inner diameter **424** of the compression mechanism **406**, shrinks into abutting contact with a portion of the lead **320** and applies enough inward force on the lead to secure the lead in place at the twist drill hole **440** so the lead is not displaced in either distal or proximal direction.

(51) The stylet **302** inserted in the lead **320** is removed from the lead in direction D. (See FIG. **10H**.) The lead **320** is left secure in place by the compression mechanism **406** and a proximal portion **328** of the lead **320** is available for routing to a neurostimulator can. (See FIG. **10I**.) A second portion **450** of the lead **320** is locked in place in a channel **414** of the lead fixation device **400** and routed along the surface of the cranium **444** to a neurostimulator can; and bone screws **408** are then used to secure the lead fixation device **400** to the surface of the cranium **444**. (See FIG. **10J**.)

(52) Thus disclosed is a lead fixation device **400, 900** for securing a first portion **448** of a lead **320** relative to a hole **440** formed through a skull. The lead fixation device **400** includes a skull attachment member **402, 902** having an upper surface **421** and a lower surface **423** and a bore **410, 910** extending through and between the upper surface and the lower surface. The lead fixation device **400, 900** also includes a lead compression mechanism **406, 906** that is integral with the skull attachment member **402, 902** and aligned with the bore **410, 910** of the skull attachment member. Integral in this context means the lead fixation device **400, 900** is formed of components assembled into a single assembly that cannot be disassembled without damaging the structural integrity of one or more of the component parts.

(53) The lead compression mechanism **406, 906** defines or forms a passageway **425, 925** through the lead fixation device **400, 900**, which passageway is characterized by a diameter **424, 924** that is defined by the lead compression mechanism. The lead compression mechanism **406, 906** is configured to transition the diameter **424, 924** from a first size to a second size greater than the first size upon insertion of an implant tool **304** through the passageway **425, 925**, and from the second size to the first size upon removal of the implant tool from the passageway.

(54) In one configuration, the compression mechanism **406, 906** is fixedly secure relative to the skull attachment member **402, 902** so that transition of the diameter **424, 924** between the first size and the second size results from radial outward compression of the compression mechanism and radial inward expansion of the compression mechanism relative to an axis **427, 927** through the passageway **425, 925**. The first portion **448** of the lead **320** has a diameter and the first size of the diameter **424, 924** of the compression mechanism **406, 906** is less than the diameter of the lead to

thereby secure the first portion of the lead in place at the hole **440**.

(55) Thus, the functionality of the lead fixation device **400, 900** is imparted by compression and expansion of the compression mechanism and no displacement or movement of the compression mechanism up or down along the axis, or about the axis is required. In fact, no relative movement between the components of the lead fixation device **400, 900** occurs when securing a lead in place—other than compression and expansion of the compression mechanism within the skull attachment member. This is distinct from prior art burr hole fixation devices where movement of one part, e.g., a lid or cap, relative to another part, e.g. a base, is required to secure a lead in place.

(56) The compression mechanism **406, 906** is formed of a flexible material, e.g., silicone, that compresses from a normal state upon receipt of force by the implant tool **304** and returns to the normal state in an absence of such force. In one configuration, the lead compression mechanism is an O-ring. In another configuration, the lead compression mechanism is formed by a pair of structures on opposite sides of the bore **410, 910** of the skull attachment member.

(57) The skull attachment member **402, 902** is formed of a material more rigid than the lead compression mechanism. The skull attachment member **402, 902** also includes at least one surface channel **414, 914** configured to receive a second portion **450** of the lead proximal the first portion of the lead.

(58) Lead Fixation Devices with Compression Mechanism Along Skull Surface

(59) With reference to FIGS. **11** and **12**, described is a first embodiment of a lead fixation device **500** configured to compress against a lead body to thereby secure the lead body relative to a skull surface adjacent a skull hole. The lead fixation device **500** prevents or at least significantly reduces movement of the lead body either further into the skull hole or out of the skull hole during implant of the lead. This lead fixation device **500** secures a portion of the lead relative to a skull surface such that the portion rests on and along a skull surface, in a parallel arrangement with the skull surface.

(60) The lead fixation device **500** includes a skull attachment member **502** configured to be secured to the skull and a flexible compression mechanism **506** that is associated with the skull attachment member. The compression mechanism **506** is configured to receive a portion of a lead body through a slot **508** and then to compress against the portion of the lead body upon receipt of a compressive force.

(61) The skull attachment member **502** is formed of a rigid plastic, e.g., PEEK. The compression mechanism **506** is formed of a material, e.g., silicone, that is softer and more flexible than the material of the skull attachment member. During production of the lead fixation device **500**, the skull attachment member **502** may be formed first with the compression mechanism **506** being formed over and within recesses of the skull attachment member to produce an integral, single piece structure with no moving parts. Integral in this context means the lead fixation device **500** is formed of components assembled into a single assembly that cannot be disassembled without damaging the structural integrity of one or more of the component parts.

(62) With reference to FIG. **12**, the compression mechanism **506** defines a passageway **504** through the skull attachment member **502**. The passageway **504** is provided by a c-shaped tube **507** having a slot **508** that extends along the length of the tube. The c-shaped tube **507** is characterized by an inner diameter **510** that is sized to receive a portion of a lead body. In one configuration, the slot **508** is characterized by an inner arc **512** that extends between 25-35 degrees around the circumference of the inside of the c-shaped tube **507**. As such, the width **514** of the slot **508** is less than the inner diameter **510** of the tube **507**.

(63) The flexibility of the c-shaped tube **507**, however, allows for the portions of the tube adjacent the slot **508** to deform upon receipt of first forces **517a, 517b** in the direction away from the axis **519** of the tube **507** and further separate from each other to thereby allow a lead body to fit through the slot **508** to the interior of the tube. Once the lead body is in the interior of the c-shaped tube **507** and the first forces **517a, 517b** are removed, the deformed portions of the tube adjacent the slot **508**

return to their pre-deformed state and the c-shaped tube assumes its normal shape to secure the lead body within the interior of the tube. In other configurations, the size of the inner arc **512** and the resulting slot **508** may be greater than or equal to the inner diameter **510** of the tube **507**.

(64) With continued reference to FIG. **12**, in one configuration the compression mechanism **506** includes a pair of wings **516a**, **516b** that extend outward from a portion of the c-shaped tube **507** in opposite directions. These wings **516a**, **516b** are formed of the same flexible material as the c-shaped tube **507** and have a thickness of about 0.5 mm. When the lead fixation device **500** is secured to the skull by bone screws, the flexibility of the pair of wings **516a**, **516b** allows them to flatten somewhat under the force of compression resulting from the screwing down of the skull attachment member **502**. This flattening translates into a receipt of second forces **518a**, **518b** in the direction toward the axis **519** of the tube **507**. These second forces **518a**, **518b** cause the width **514** of the slot **508** in the area of the wings **516a**, **516b** to reduce, which serves to further secure the lead in place and prevent movement of the lead body in either direction along the length of the tube **507**. Compression of the silicone c-shaped tube **507** itself along the length of the tube may also serve to secure the lead in place.

(65) With reference to FIG. **11**, the compression mechanism **506** may also include a pair of circumferential grooves **520a**, **520b** configured to secure a suture in place around the c-shaped tube **507**. Sutures may thus be applied at either end of the c-shaped tube to further secure the lead body in place. However, given the above described functionality of the wings **516a**, **516b**, the use of sutures is optional as the second forces **518a**, **518b** imparted by the wings by the securing of the device to the skull is sufficient to secure the lead in place and prevent movement of the lead body in either direction along the length of the compression mechanism **506**.

(66) With continued reference to FIG. **11**, the skull attachment member **502** includes a pair of screw holes **522a**, **522b** configured to receive a bone screw. The skull attachment member **502** has a maximum thickness of 2 mm in the region of the screw holes and tapers downward to a reduced thickness at the side edges.

(67) With reference to FIG. **13**, an example implant procedure for the lead fixation device of FIGS. **11** and **12** is provided. A twist drill hole **530** is formed in location of the cranium using known stereotactic techniques. A lead **532** is implanted using known techniques.

(68) The passageway **504** of the compression mechanism **506**, as defined by a c-shaped tube **507**, is aligned with the body of the lead **532** and a portion of the lead body is pushed through the slot **508** into the passageway of the c-shaped tube. The lower surface **538** of the skull attachment member **502** is positioned relative to the skull surface to place the wings **516a**, **516b** of the compression mechanism **506** on the skull surface.

(69) The lead fixation device **500** is secured to the surface **544** skull using bone screws **524**. The screws are advanced into the bone to a level sufficient to cause the pair of wings **516a**, **516b** to flatted and apply second forces **518a**, **518b** that cause the width **514** of the slot **508** in the area of the wings **516a**, **516b** to reduce, which serves to secure the lead in place and prevent movement of the lead body in either direction along the length of the tube.

(70) Thus disclosed is a lead fixation device **500** for securing a portion **534** of a lead **532** relative to a surface **544** of a skull. The lead fixation device **500** includes a skull attachment member **502** having an upper surface **536** and a lower surface **538**. The lead fixation device **500** also includes a lead compression mechanism **506** that is integral with the skull attachment member **502**. Integral in this context means the lead fixation device **500** is formed of components assembled into a single assembly that cannot be disassembled without damaging the structural integrity of one or more of the component parts. The lead compression mechanism **506** is formed of a flexible material, e.g., silicone, while the skull attachment member **502** is formed of a material more rigid than the lead compression mechanism, such as PEEK.

(71) The lead compression mechanism **506** forms a passageway **504** through the skull attachment member **502**. The passageway **504** is characterized by a slot **508** that faces in the direction of the

lower surface **538** and extends along the length of the passageway.

(72) The lead compression mechanism **506** is formed of a flexible material, e.g., silicone, and is configured to transition a width **514** of the slot **508** from an initial size that is less than the diameter of the portion **534** of the lead **532**, to an expanded size that is greater than the initial size upon receipt of appropriately directed first forces **517a**, **517b** in a region of the lead compression mechanism **506** that forms the passageway **504**. These first forces **517a**, **517b** may be respectively applied against a facing edge of the c-shaped tube **507** in opposite direction relative to the axis **519** of the passageway **504**. As such, the first forces **517a**, **517b** increase the width **514** and allow for the portion **534** of the lead **532** to slide into the passageway **504**.

(73) The lead compression mechanism **506** is further configured to transition the width **514** of the slot **508** from the initial size to a collapsed size upon receipt of appropriately directed second force **518a**, **518b** in a region of the lead compression mechanism **506** that forms the passageway **504**. These second forces **518a**, **518b** may be respectively applied along an outer surface of the c-shaped tube **507** toward each other. As such, the second forces **518a**, **518b** decrease the width **514** and secure the portion **534** of the lead **532** inside the passageway **504**. In one configuration, the lead compression mechanism **506** includes at least one wing **516a**, **516b** that extends to a side of the passageway **504** and along the lower surface **538** of the skull attachment member **502**. These wings **516a**, **516b** function to distribute the second forces **518a**, **518b** along the c-shaped tube **507**.

(74) With reference to FIGS. **14**, **15A**, and **15B**, described is a second embodiment of a lead fixation device **600** configured to compress against a lead body to thereby secure the lead body relative to a skull surface adjacent a skull hole and prevent or at least significantly reduce movement of the lead body either further into the skull hole or out of the skull hole during implant of the lead. This lead fixation device **600**, like the fixation device **500** described above with reference to FIGS. **11** and **12**, secures a portion of the lead relative to a skull surface such that the portion rests on and along a skull surface, in a parallel arrangement with the skull surface.

(75) The lead fixation device **600** includes a skull attachment member **602** configured to be secured to the skull and a flexible compression mechanism **606** that is associated with the skull attachment member. The compression mechanism **606** is configured to receive a portion of a lead body through a slot **608** and then to compress against the portion of the lead body upon receipt of a compressive force.

(76) The skull attachment member **602** is formed of a rigid plastic, e.g., PEEK. The compression mechanism **606** is formed of a material, e.g., silicone, that is softer and more flexible than the material of the skull attachment member. During production of the lead fixation device **600**, the skull attachment member **602** may be formed first, with the compression mechanism **606** being formed over and within recesses of the skull attachment member to produce an integral, single-piece structure. Integral in this context means the lead fixation device **600** is formed of components assembled into a single assembly that cannot be disassembled without damaging the structural integrity of one or more of the component parts.

(77) With reference to FIG. **14**, the compression mechanism **606** defines a passageway **604** through the skull attachment member **602**. The passageway **604** is provided by a c-shaped tube **607** having a slot **608** that extends along the length of the tube. The c-shaped tube **607** is configured the same as the c-shaped tube **507** described above with reference to FIGS. **11** and **12**. Thus, the c-shaped tube **607** is characterized by an inner diameter **610** that is sized to receive a lead body. The slot **608** is characterized by an inner arc **613** that extends between 25-35 degrees around the circumference of the inside of the c-shaped tube **607**. As such, the width **614** of the slot **608** is less than the inner diameter of the tube **607**.

(78) The flexibility of the c-shaped tube **607**, however, allows for the portions of the tube adjacent the slot **608** to deform upon receipt of first forces **617a**, **617b** and further separate from each other to thereby allow a portion of a lead body to fit through the slot to the interior of the tube. Once the portion of the lead body is within the interior of the c-shaped tube **607** and the first forces **617a**,

617b are removed, the deformed portions of the tube adjacent the slot **608** return to their pre-deformed state and the c-shaped tube assumes its normal shape to secure the lead body within the interior of the tube.

(79) The compression mechanism **606** may also include a pair of circumferential grooves **620a**, **620b** configured to secure a suture in place around the c-shaped tube **607**. Sutures may thus be applied at either end of the c-shaped tube **607** to further secure the lead body in place and prevent movement of the lead body in either direction along the length of the tube. The skull attachment member **602** includes a screw hole configured to receive a bone screw **612**.

(80) With reference to FIGS. **16A** and **16B**, an example implant procedure for the lead fixation device of FIGS. **14**, **15A**, and **15B** is provided. A twist drill hole **630** is formed in location of the cranium **644** using known stereotactic techniques. A lead **632** is implanted using known techniques such that a portion of the lead passes through the twist drill hole **630** and is beneath the cranium **644** in the brain, and a portion of the lead rests on top of the cranium.

(81) The compression mechanism **606** of a lead fixation device **600** is aligned with the body of the lead **632** and a portion **634** of the lead body is pushed through the slot **608** into the compression mechanism. The skull attachment member **602** of the lead fixation device **600** is positioned relative to the surface of the cranium **644** and is secured to the skull using a bone screw **612**. Sutures are secured around the pair of circumferential grooves **620a**, **620b** to secure the portion **634** of the lead **632** body in place in the compression mechanism **606** and prevent movement of the lead body in either direction along the length of the compression mechanism.

(82) Thus disclosed is a lead fixation device **600** for securing a portion **634** of a lead **632** relative to a surface of a skull or cranium **644**. The lead fixation device **600** includes a skull attachment member **602** having an upper surface **636** and a lower surface **638**. The lead fixation device **600** also includes a lead compression mechanism **606** that is integral with the skull attachment member **602**. Integral in this context means the lead fixation device **600** is formed of components assembled into a single assembly that cannot be disassembled without damaging the structural integrity of one or more of the component parts. The lead compression mechanism **606** is formed of a flexible material, e.g., silicone, while the skull attachment member **602** is formed of a material more rigid than the lead compression mechanism, such as PEEK.

(83) The lead compression mechanism **606** forms a passageway **604** through the skull attachment member **602**. The passageway **604** is characterized by a slot **608** that faces in the direction of the upper surface **636** and extends along the length of the passageway.

(84) The lead compression mechanism **606** is formed of a flexible material, e.g., silicone, and is configured to transition a width **614** of the slot **608** from an initial size that is less than the diameter of the portion **634** of the lead **632**, to an expanded size that is greater than the initial size upon receipt of appropriately directed first forces **617a**, **617b** in a region of the lead compression mechanism **606** that forms the passageway **604**. These first forces **617a**, **617b** may be respectively applied against a facing edge of the c-shaped tube **607** in opposite direction relative to the axis **619** of the passageway **604**. As such, the first forces **617a**, **617b** increase the width **614** and allow for the portion **634** of the lead **632** to slide into the passageway **604**.

(85) The lead compression mechanism **606** is further configured to transition the width **614** of the slot **608** from the initial size to a collapsed size upon receipt of appropriately directed second force **618a**, **618b** in a region of the lead compression mechanism **606** that forms the passageway **604**. These second forces **618a**, **618b** may be respectively applied along an outer surface of the c-shaped tube **607** toward each other. As such, the second forces **618a**, **618b** decrease the width **614** and secure the portion **634** of the lead **632** inside the passageway **604**. The second forces **618a**, **618b** may result from the placement and tying of a suture in each of the grooves **620a**, **620b**.

(86) The various aspects of this disclosure are provided to enable one of ordinary skill in the art to practice the present invention. Various modifications to exemplary embodiments presented throughout this disclosure will be readily apparent to those skilled in the art. Thus, the claims are

not intended to be limited to the various aspects of this disclosure, but are to be accorded the full scope consistent with the language of the claims. All structural and functional equivalents to the various components of the exemplary embodiments described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

Claims

1. A lead fixation device for securing a portion of a lead relative to a surface of a skull, the lead fixation device comprising: a skull attachment member having an upper surface and a lower surface; and a compression mechanism integral with the skull attachment member and forming a passageway relative to the skull attachment member, which passageway is characterized by a slot that extends along a length of the passageway, wherein the compression mechanism is configured to: transition a width of the slot from an initial size to an expanded size that is greater than the initial size upon a receipt of appropriately directed first forces in a region of the compression mechanism that forms the passageway, and transition the width of the slot from the initial size to a collapsed size upon a receipt of appropriately directed second forces in a region of the compression mechanism that forms the passageway.
 2. The lead fixation device of claim 1, wherein the compression mechanism is formed of a flexible material.
 3. The lead fixation device of claim 1, wherein the portion of the lead has a diameter, and the initial size is less than the diameter and the expanded size is greater than the diameter.
 4. The lead fixation device of claim 1, wherein the skull attachment member is formed of a material more rigid than the compression mechanism.
 5. The lead fixation device of claim 1, wherein the slot faces in the direction of the lower surface of the skull attachment member.
 6. A lead fixation device for securing a portion of a lead relative to a surface of a skull, the lead fixation device comprising: a skull attachment member having an upper surface and a lower surface; and a compression mechanism integral with the skull attachment member and forming a passageway relative to the skull attachment member, which passageway is characterized by a slot that extends along a length of the passageway and faces in the direction of the lower surface of the skull attachment member, wherein the compression mechanism comprises at least one wing extending to a side of the passageway and along the lower surface of the skull attachment member and is configured to: transition a width of the slot from an initial size to an expanded size that is greater than the initial size upon a receipt of appropriately directed first forces in a region of the compression mechanism that forms the passageway, and transition the width of the slot from the initial size to a collapsed size upon a receipt of appropriately directed second forces in a region of the compression mechanism that forms the passageway.
 7. The lead fixation device of claim 1, wherein the slot faces in the direction of the upper surface of the skull attachment member.
 8. The lead fixation device of claim 1, wherein the passageway extends along the upper surface of the skull attachment member.
 9. The lead fixation device of claim 1, wherein the passageway extends along the lower surface of the skull attachment member.
-